

Hyesung Kim

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Bundang-gu, Seungnam-si, Gyeonggi-do, Republic of Korea

Research Interest

- **Multimodal:** Multimodal Representation Learning, Multimodal Alignment
- **NLP:** Large Language Model
- **Computer Vision:** multimodal interaction, Visual-Language Model

Education

Sungkyunkwan University, BS in Computer Education Mar 2020 – Present

- GPA: 4.02/4.5
- **Coursework:** Introduction to Deep Neural Networks, Artificial Intelligence, Natural Language Processing, Linear Algebra

Experience

HLILab Summer Intern Jul 2025 – Present

- Working on voice phishing detection

Teaching Practicum, Bundang High School May 2025

- Delivered lessons in Informatics to high school students under the supervision of mentor teacher
- Observed classroom management strategies and assisted with student counseling

LG Aimers 6th Cohort Jan 2025 – Feb 2025

- Competed in a hackathon to predict pregnancy success rates for patients with infertility, achieving a top 6% ranking on the leaderboard

Google Machine Learning Bootcamp 2024 Jul 2024 – Oct 2024

- Completed the "Deep Learning Specialization" course series by Andrew Ng on Coursera
- Developed a Korean psychological counseling chatbot by fine-tuning the Gemma2-2B-IT model with PEFT techniques during the Gemma Sprint Project

Republic of Korea Air Force, System Development Unit Mar 2022 – Dec 2023

- Developed an air-to-ground shooting score evaluation program integrating object detection and semantic segmentation techniques

Projects

Undergraduate Researcher Project, Sungkyunkwan University Sep 2024 – Dec 2024

- Developed an emotional dialogue chatbot by fine-tuning BlenderBot-small on the ESConv dataset
- Created an interactive web application for real-time chatting using React and Flask
- Advisor: Prof. Jinyoung Han, Department of Applied Artificial Intelligence

AI Capstone Design Project Mar 2024 - Jun 2024

- Worked on anomaly detection in respiratory sounds by extracting Mel-Spectrogram images and MFCC, Chroma, and Poly features
- Built a multi-input classification model based on ResNet-18 to classify respiratory anomalies

Wasabi Warriors – Artificial Intelligence Course, Sungkyunkwan University May 2024 - Jun 2024

- Collaborated with a team of three to perform multi-class classification and object detection on sushi images

- Applied transfer learning techniques and evaluated model performance using standard metrics (accuracy, mAP)

MusicNet – CMD, Computer Education Society, Sungkyunkwan University

Sep 2021 - Dec 2021

- Collaborated with four team members from the CMD
- Preprocessed jazz MIDI files by separating sessions and developed a jazz music generation model using MuseGAN
- Awarded the Popularity Prize at the AI x ART Competition (National Science Museum, Korea) for AI-based jazz music generation

Awards and Certificates

- Academic Excellence Scholarship, Project Group for Education and Research in Medical AI – Feb 2025
- SKKU Academic Excellence Scholarship – Feb 2025
- Special Award, K-PaaS Service Development and Idea Contest (hosted by MSIT, NIA) – Dec 2024
- Technology Award, 2024 CO-SHOW x AIM Challenge (hosted by AI COSS) – Nov 2024
- Encouragement Award, 2024 Capstone Design and AI Hackathon (hosted by Korean Association of Computer Education) – Oct 2024
- Encouragement Award, 6th Seoul Education Data Analysis and Utilization Contest (hosted by Seoul Metropolitan Office of Education) – Sep 2024
- Excellence Award in AI Program Category, 5th Air Force Innovation Hackathon – Nov 2023
- Engineer Information Processing - Jun 2025
- ADSP (Advanced Data Analytics Semi-Professional Certification) – Sep 2024
- Industrial Engineer Information Processing – Nov 2023
- SQLD (SQL Developer Certification) – Sep 2023

Skills

Language: Korean(Native), English(Fluent)

Programming: Python, PyTorch, JavaScript, C, React Native